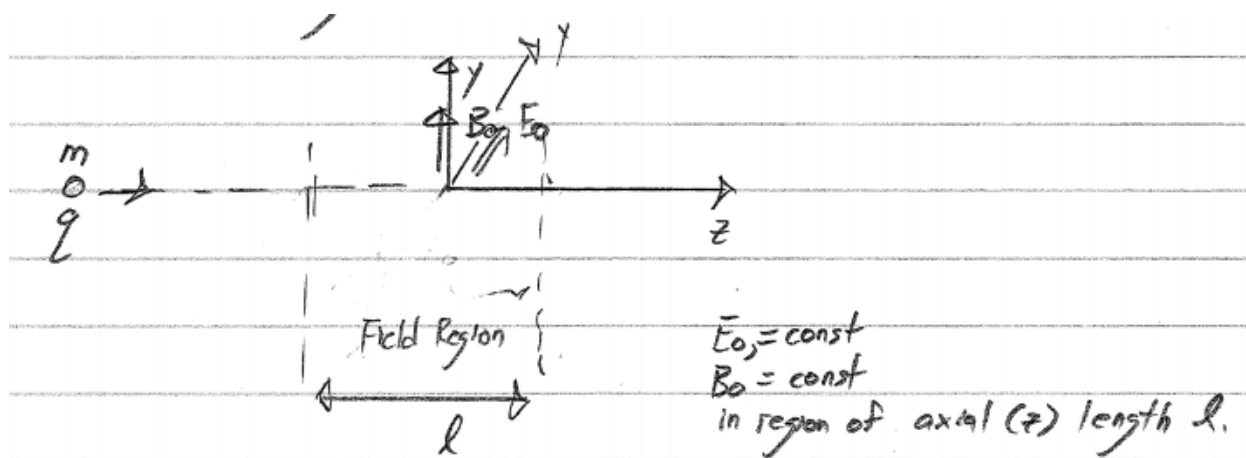


Instructions:

- Solutions are due by 9:00 a.m. the next morning.
- While points may differ between Problem Sets each Problem Set will carry equal weight for course grade.
- Please list your collaborators on your submission.

Problem 1. Charge to Mass Selector 30 pts.

Positively charged ions with charge q and mass m are emitted from a source and accelerated through a potential difference V_0 . The particles then enter a “velocity selector” made up of crossed electric and magnetic fields, E_0 and B_0 . Model these fields as uniform over an axial length, ℓ .



- 5 pts: Find a formula for the value of axial β such that the particle will move through the selector without deflection.
- 3 pts: What value of β will pass through the selector undeflected for $E_0 = 75 \text{ kV/m}$ and $B_0 = 0.05 \text{ T}$?
- 5 pts: Assume non-relativistic motion. Show that if the magnetic field is turned off ($B_0 = 0$) that particles are deflected through an angle θ emerging from the deflector, with θ satisfying:

$$\tan(\theta) = \frac{1}{2} \frac{E_0 \ell}{V_0}$$

- 2 pts: If $\ell = 0.15 \text{ m}$, $V_0 = 60 \text{ kV}$, and $E_0 = 75 \text{ kV/m}$ evaluate θ in part (c).
- 5 pts: Assume the electric field E_0 and magnetic field B_0 are in balance for no deflection of the particles. Then the electric field is set to zero ($E_0 = 0$) and the magnetic field is left at its original value B_0 . Show that the particles will be bent on a circular arc of radius:

$$R = \frac{2V_0}{E_0}$$

- 2 pts: Find the value of R for $V_0 = 60 \text{ kV}$ and $E_0 = 75 \text{ kV/m}$.

- g) 3 pts: Show for undeflected particles that the charge-to-mass ratio is:

$$\frac{q}{m} = \frac{E_0^2}{2V_0 B_0^2}$$

If ions have charge state $Q = q/e$ and atomic mass $A = m/m_u$, where $m_u = 931 \text{ MeV}/c^2$ is the rest mass of a nucleon, this value of q/m corresponds to:

$$\frac{Q}{A} = \frac{m_u c^2}{2eV_0} \left(\frac{E_0}{cB_0} \right)^2 = \frac{m_u c^2}{2eV_0} \beta^2$$

- h) 5 pts: Use the results and parameters above to find the value of Q/A for undeflected ions. If we believe those ions to be nitrogen, what charge state is most likely?

Charge to Mass Selector 30 pts.

- a) 5 pts:

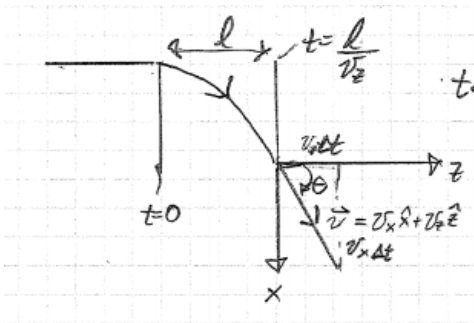
$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

$$E_0 \mathbf{x} - \beta c B_0 = 0$$

$$\beta = \frac{E_0}{cB_0}$$

- b) 3 pts: $\beta = 0.005$

- c) 5 pts:



Calculate the x velocity at the exit:

$$F_x = qE_0 = m \frac{d^2 x}{dt^2}$$

$$\dot{x}(t) = \frac{qE_0}{m} t + \text{constant}$$

$$v_x|_{\text{exit}} = \frac{qE_0}{m} \frac{\ell}{v_z}$$

v_z is a constant because there is no longitudinal force therefore:

$$\tan(\theta) = \frac{v_x}{v_z} \Big|_{\text{exit}} = \frac{qE_0}{m} \frac{\ell}{v_z^2}$$

But $1/2mv_z^2 = qV$ due to energy conservation

$$\Rightarrow \boxed{\tan(\theta) = \frac{E_0 \ell}{2V}}$$

d) 2 pts: $\boxed{\theta = 93.5 \text{ mrad} = 5.38^\circ}$

e) 5 pts: From part a):

$$\beta = \frac{E_0}{cB_0}, \quad v = E_0 B_0$$

Circular arc in the magnetic field when the electric field is zero:

$$m \frac{v^2}{R} = qvB_0 \Rightarrow R = \frac{mv}{qB_0} = \frac{mv^2}{qE_0}$$

But energy is constant:

$$\frac{1}{2}mv^2 = qV_0$$

$$\Rightarrow \boxed{R = 2 \frac{V_0}{E_0}}$$

f) 2 pts: $\boxed{R = 1.6 \text{ m}}$

g) 3 pts: Undeflected $v = E_0/B_0$ from part a). By energy conservation:

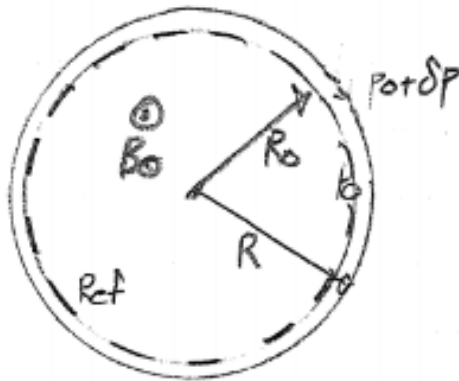
$$\frac{1}{2}mv^2 = qV_0 \Rightarrow \boxed{\frac{q}{m} = \frac{1}{2} \frac{v^2}{V_0} = \frac{E_0^2}{2B_0^2 V_0}}$$

h) 5 pts: $N \Rightarrow A = 14$, $m_u c^2 = 931 \text{ MeV}$, $\beta = 0.005$ from part b), $eV_0 = 60 \text{ keV}$.

$Q/A = 0.194$. $A = 14 \Rightarrow Q = 2.72$. Probable charge state $\boxed{Q = 3}$.

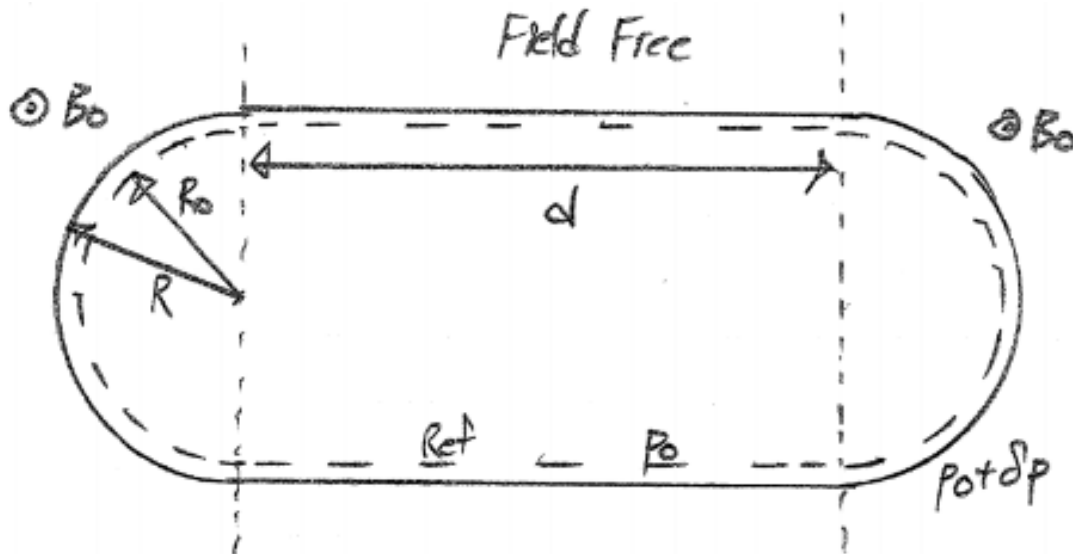
Problem 2. Slip Factor 20 pts.

- a) 5 pts: Consider a circular accelerator/storage ring composed of a uniform magnetic field, $\mathbf{B}_y = B_0 \hat{y} = \text{constant}$. The ideal reference path for a particle of momentum p_0 has radius R_0 in the plane perpendicular to $\mathbf{B}_y$. A particle with momentum $p = p_0 + \delta p$ will have a different closed path and radius R .



Calculate the slip factor, η , in terms of γ for this situation.

- b) 5 pts: Next, repeat part (a) for a racetrack accelerator with two uniform dipole bends separated by a field free drift of length d . Calculate the slip factor, η , in terms of γ and d/R_0 .



- c) 10 pts: For $d = 2R_0$ in part (b), plot η as a function of γ and note where it changes sign. Is this the “transition γ ”? What speed in $\beta = v/c$ does this correspond to?

SOLUTION: Slip Factor 20 pts.

a) 5 pts:

$$\begin{aligned}
-\eta_s &= \alpha_c - \frac{1}{\gamma^2} \\
\alpha_c &= \frac{\Delta L/L_0}{\delta p/p_0} \\
L &= 2\pi R_0, \quad \Delta L = 2\pi(R - R_0), \implies \frac{\Delta L}{L} = \frac{R - R_0}{R} \\
[B\rho] &= \frac{p}{q}, \quad \frac{1}{R} = \frac{B_0}{[B\rho]} = \frac{qB_0}{p} \\
p_0 &= qB_0R_0, \quad p = qB_0R = p_0 + \delta p \\
\implies \delta p &= qB_0(R - R_0), \quad \frac{\delta p}{p_0} = \frac{R - R_0}{R_0} \\
-\eta_s &= \frac{(R - R_0)/R_0}{(R - R_0)/R_0} - \frac{1}{\gamma^2} = 1 - \frac{1}{\gamma^2}
\end{aligned}$$

b) 5 pts: In this case:

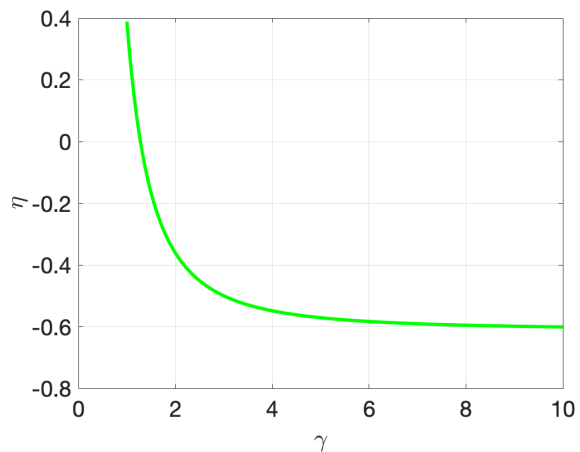
$$\begin{aligned}
L_0 &= 2\pi R_0 + 2d, \quad L = 2\pi R + 2d, \quad \Delta L = 2\pi(R - R_0) \\
\therefore \frac{\Delta L}{L} &= \frac{2\pi(R - R_0)}{2\pi R_0 + 2d} \\
\frac{\delta p}{p_0} &= \frac{R - R_0}{R_0} \\
-\eta_s &= \frac{\pi(R - R_0)/(\pi R_0 + d)}{(R - R_0)/R_0} - \gamma^2 \\
-\eta_s &= \frac{\pi}{\pi + d/R_0} - \frac{1}{\gamma^2}
\end{aligned}$$

c) 10 pts:

$$d = 2R_0 \implies -\eta_s = \frac{\pi}{\pi + 2} - \frac{1}{\gamma^2}$$

At the transition by definition $\eta_s = 0$, $\gamma = \gamma_t$

$$\begin{aligned}
\gamma_t &= \sqrt{\frac{\pi + 2}{\pi}} \simeq 1.279 \\
\beta_t &= \sqrt{1 - \frac{1}{\gamma_t^2}} \simeq 0.624
\end{aligned}$$

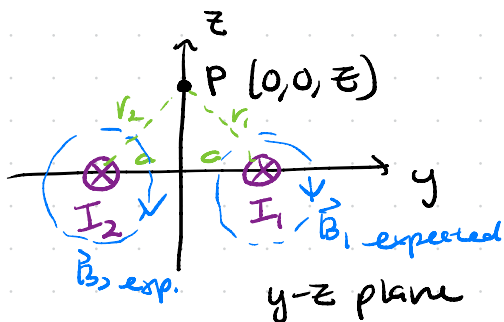
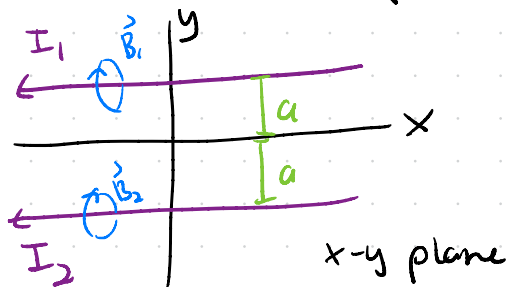


Problem 3. Magnetic field from two straight wires 30 pts.

Two infinitely long, straight wires each carry a current I in the $-x$ direction. The wires are separated by a distance $2a$ in the y direction.

- (a) 15 pts: Find an expression for the total magnetic field from the two wires at a point on the z axis halfway between the two wires, $(0, 0, z)$.
- (b) 10 pts: Find the $z > 0$ value where the field is maximized.
- (c) 5 pts: Draw a sketch of the field lines in the y - z plane for the magnetic field produced by the two wires.

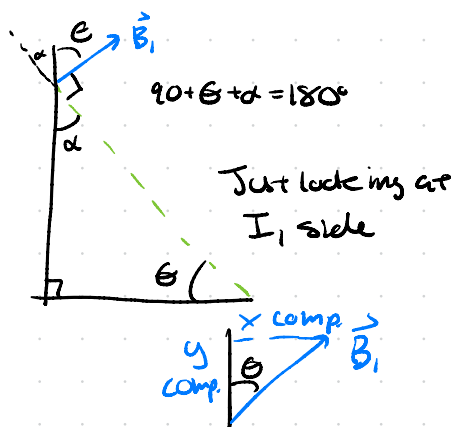
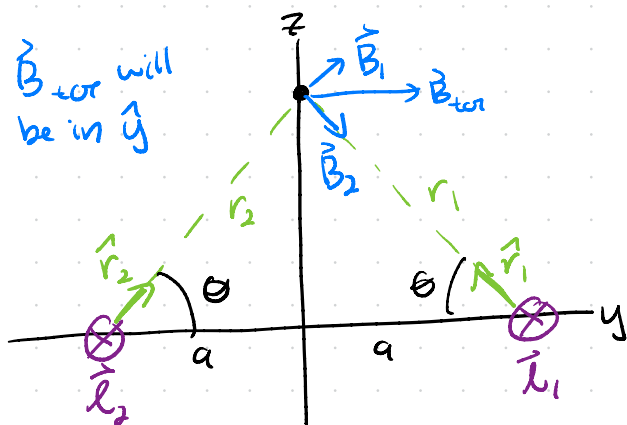
(a) Sketching the problem:



The magnitude of each field at P is easy to find from Ampere's law/what we know about the field of a long wire:

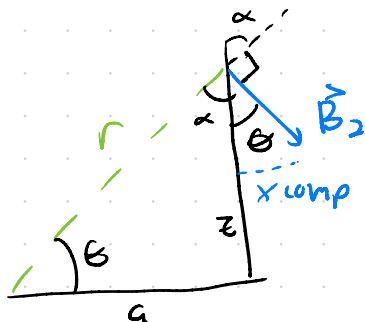
$$r_1 = r_2 = \sqrt{a^2 + z^2} \quad B(r) = \frac{\mu_0 I}{2\pi r} \Rightarrow B(P) = \frac{\mu_0 I}{2\pi \sqrt{a^2 + z^2}} = B_1 = B_2$$

But we also need to find the direction: (Biot-Savart)



Can see from the sketch above $|\vec{B}_1| = B_1$ we find

$$\vec{B}_1 = B_1 (\sin \theta \hat{y} + \cos \theta \hat{z}) \text{ and the same analysis for } \vec{B}_2 \text{ gives: } \vec{B}_2 = B_2 (\sin \theta \hat{y} - \cos \theta \hat{z})$$



* Both B_1 and B_2
are same $+\hat{y}$, opposite
in $\hat{z}$

$$\vec{B} = \vec{B}_1 + \vec{B}_2 = \frac{\mu_0 I}{2\pi\sqrt{a^2+z^2}} (2\sin\theta \hat{y}) \quad \sin\theta = \frac{z}{\sqrt{a^2+z^2}} \quad \checkmark$$

plug in $\sin\theta \Rightarrow$ $\boxed{\vec{B} = \frac{\mu_0 I z}{\pi(a^2+z^2)} \hat{y}}$ at P, but valid any point along z (watch sign of z will give correct direction)

(b) To find z with max $B(z)$, take dB/dz and find zeroes.

Use chain rule: $\frac{d}{dx}(u/v) = \frac{vu' - uv'}{v^2}$

$u = \mu_0 I z, \quad v = \pi(a^2+z^2), \quad u' = \mu_0 I, \quad v' = 2\pi z$

$$\frac{dB}{dz} = \frac{\cancel{\pi}(a^2+z^2)(\mu_0 I) - \mu_0 I z(2\cancel{\pi}z)}{(\pi(a^2+z^2))^2} = \frac{\mu_0 I(a^2 + \cancel{z^2} - 2\cancel{z^2})}{\pi(a^2+z^2)^2}$$

$$\frac{dB}{dz} = \frac{\mu_0 I}{\pi} \frac{a^2 - z^2}{(a^2+z^2)^2} \quad \text{Want } \frac{dB}{dz} = 0 = \frac{a^2 - z^2}{(a^2+z^2)^2}$$

so $a^2 - z^2 = 0 \Rightarrow \boxed{z = a}$

③ Close to wires: field from wire.
Between / farther away, they add: $\pm \hat{y}$ on
 z axis, $\pm \hat{z}$ on y axis

